

Monty Hall

puerta

$X: (\Omega, \mathcal{P}) \rightarrow \{1, 2, 3\}$ Dónde está el coche

$Y: (\Omega, \mathcal{P}) \rightarrow \{1, 2, 3\}$ puerta que elige el concursante al comenzar

$M: (\Omega, \mathcal{P}) \rightarrow \{\text{Cabra}, \text{Cochete}\}$ lo que se encuentra detrás de la puerta que el moderador escoge

$$\Rightarrow P(M = \text{Cabra}) = \frac{1}{2}$$

$P(X=Y) = 1/3$ Probabilidad de seleccionar la puerta correcta

$$P(M = \text{Cabra} | X=Y) = P(M = \text{Cabra} | X \neq Y) = 1$$

$$P(X=Y | M = \text{Cabra}) = \frac{P(X=Y, M = \text{Cabra})}{P(M = \text{Cabra})} = 1$$

$$= P(M = \text{Cabra} | X=Y) \cdot P(X=Y) = 1/3$$

$$P(X \neq Y | M = \text{Cabra}) = 1 - P(X=Y | M = \text{Cabra}) = 2/3$$

Monty Hall Randomly (Moderador no sabe dónde está el coche)

$$\begin{aligned} \text{Entonces: } P(M = \text{Cabra}) &= P(M = \text{Cabra} | X=Y) \cdot P(X=Y) + P(M = Y | X \neq Y) \cdot P(X \neq Y) \\ &= 1 \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{3} \end{aligned}$$

$$\text{Luego } P(X=Y | M = \text{Cabra}) = \frac{P(M = \text{Cabra} | X=Y) \cdot P(X=Y)}{P(M = \text{Cabra})} = \frac{1/3}{2/3} = 1/2$$

Análisis 1º Curso Grado $n=100$ | $P(X=y | M=\omega_n) = P(X=y) = \frac{1}{n}$

P2) $X|\theta \sim B_{1,n}(n, \theta)$ $n \in \mathbb{N}$
 $\theta | \alpha, \beta \sim \text{Beta}(\alpha, \beta)$ $\alpha, \beta \in \mathbb{N}^+$

$$f_{X|\theta}(x) = \binom{n}{x} \theta^x (1-\theta)^{n-x}, \quad f_{\theta}(\theta) = \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

a) $h(x, \theta) = p(X|\theta) \cdot p(\theta)$

$$= \binom{n}{x} \theta^x (1-\theta)^{n-x} \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

$$= \frac{\binom{n}{x}}{B(\alpha, \beta)} \theta^{x+\alpha-1} (1-\theta)^{n+\beta-x-1}$$

b) $m(x) = \int_{\theta} h(x, \theta) d\theta = \frac{\binom{n}{x}}{B(\alpha, \beta)} \int_{\theta} \frac{\theta^{x+\alpha-1} (1-\theta)^{n+\beta-x-1}}{B(x+\alpha, n+\beta-x)} d\theta$

$$= \binom{n}{x} \cdot \frac{\Gamma(\alpha)\Gamma(\beta) \cdot \Gamma(n+\alpha+\beta)}{\Gamma(\alpha+\beta) \Gamma(x+\alpha) \Gamma(n+\beta-x)}$$

c) $\pi(\theta|x) = \frac{p(x|\theta) p(\theta)}{m(x)} = \frac{\binom{n}{x} \theta^x (1-\theta)^{n-x} \cdot \theta^{\alpha-1} (1-\theta)^{\beta-1} \cdot 1}{\binom{n}{x} \frac{B(x+\alpha, n-x+\beta)}{B(\alpha, \beta)}} = \frac{\theta^{x+\alpha-1} (1-\theta)^{n-x+\beta-1}}{B(x+\alpha, n-x+\beta)}$

$\rightarrow \theta|x \sim \text{Beta}(x+\alpha, n-x+\beta)$

or $\pi(\theta|x) \propto p(x|\theta) p(\theta) \propto \theta^{x+\alpha-1} (1-\theta)^{n-x+\beta-1}$
 $\rightarrow \theta|x \sim \text{Beta}$

d) $m(x_0|x) = \int_{\theta} p(x_0|\theta) d\pi(\theta|x)$

Assuming $x_0 \perp x | \theta$
 ind. conditions

$$= \int_0^1 \binom{n}{x_0} \theta^{x_0} (1-\theta)^{n-x_0} \cdot \frac{\theta^{x+\alpha-1} (1-\theta)^{n-x+\beta-1}}{B(x+\alpha, n-x+\beta)} d\theta$$

$$= \frac{\binom{n}{x_0} B(x_0+x+\alpha, 2n-x-x_0+\beta)}{B(n+2, n-x+\beta)} \cdot \frac{B(x+\alpha, n-x+\beta)}{B(x_0+x+\alpha, 2n-x-x_0+\beta)}$$

1

(2)

P3)

$$Y|n, p \sim \text{Bin}(n, p)$$

$$p \text{ constant} \rightarrow p(n) = \frac{m^n e^{-m}}{n!}$$

$$n \sim \text{Poisson}(m)$$

$$p(n|Y) \propto p(Y|n) \cdot p(n)$$

$$\binom{n}{Y} p^Y (1-p)^{n-Y} \cdot \frac{m^n e^{-m}}{n!}$$

$$= \frac{n!}{(n-Y)! Y!} \cdot \frac{1}{n!} \frac{p^Y}{(1-p)^Y} [(1-p)m]^n \cdot e^{-m}$$

$$\propto \frac{[(1-p)m]^{n-Y}}{(n-Y)!} \cdot \frac{[(1-p)m]^Y}{[(1-p)m]^Y} \cdot e^{-(1-p)m}$$

$$\propto \frac{[(1-p)m]^{n-Y}}{(n-Y)!} e^{-(1-p)m}$$

$$\rightarrow n-Y|Y \sim \text{Poisson}((1-p)m)$$

$$P(n \leq x|Y) = P(n-Y \leq x-Y|Y) \stackrel{\sim \text{Poisson}}{=} \frac{[(1-p)m]^{x-Y}}{(x-Y)!}$$

$$\boxed{n \geq Y}$$

$$A = n-Y \sim \text{Poisson} \rightarrow n = A+Y$$

b) Given m and p poisson A given the sum Y .